

VYVAR Install (EN)

VYVAR release bundle - English install guide (RELEASE-2)

Preview bundles ship a **bundled Python 3.12 runtime** (R1). Your system Python is not used. User data (config, database, Archive, catalogs) lives in a separate **data directory** from the install folder.

1. Download and unpack

1. Download `VYVAR--win64.zip` or `VYVAR--linux-x64.tar.gz` from GitHub Releases on the public `VYVAR-release` repository.
2. Verify SHA256 against `SHA256SUMS` in the same release.
3. Unpack to a permanent location, e.g.:
 - Windows: `C:\Program Files\VYVAR\`
 - Linux: `~/apps/vyvar/`
4. Paths with spaces are supported.

2. First launch

Windows: double-click `VYVAR.bat` or run from cmd:

```
...  
cd C:\Program Files\VYVAR\VYVAR-preview-20260723-win64  
VYVAR.bat  
...
```

Linux:

```
...  
chmod +x vyvar.sh  
./vyvar.sh  
...
```

On first launch the data directory is created automatically:

Platform Default data directory

Windows `%LOCALAPPDATA%\VYVAR`

Linux `~/local/share/vyvar`

Override before launch: set environment variable `VYVAR\_DATA\_DIR` to any writable path.

The initializer creates `Archive/Drafts/`, `CalibrationLibrary/`, catalog folders, `logs/`, an empty `vyvar.sqlite3`, `config.json` from template, and `NEXT\_STEPS.txt`.

Catalog and observation data never live inside the install folder (B1 design).

3. Health check (selftest)

```

...
VYVAR.bat --selftest
./vyvar.sh --selftest
...

```

Prints Python version, platform, data directory, data-dir skeleton status, key dependency versions, runtime file checks, and imports every compiled science module. Exit code 0 means the install is healthy.

4. Building the catalogs (required - never pre-built)

Catalogs are **always** built by the user (R2). The release bundle ships builder scripts under ``scripts/catalogs/`` (same bundled Python; no system Python).

4.1 Order (do not skip Gaia)

1. **Gaia DR3 SQLite** (largest; network download from ESA Gaia TAP)
2. **Gaia blind indexes** (local CPU; requires Gaia DB from step 1)
3. **VSX local DB** (VizieR download)
4. **Exoplanet local DB** (NASA Exoplanet Archive TAP)

4.2 Commands (bundled launcher)

Use ``--`` before script flags. Outputs default to your **data directory** (section 2).

Linux:

```

...
./vyvar.sh --tool build_gaia -- --help
./vyvar.sh --tool build_gaia -- --mag-limit 16.5
./vyvar.sh --tool build_blind_index --
./vyvar.sh --tool build_vsx --
./vyvar.sh --tool build_exoplanets --
...

```

Windows:

```

...
VYVAR.bat --tool build_gaia -- --help
VYVAR.bat --tool build_gaia -- --mag-limit 16.5
VYVAR.bat --tool build_blind_index --
VYVAR.bat --tool build_vsx --
VYVAR.bat --tool build_exoplanets --
...

```

See also ``scripts/catalogs/README.md`` in the install folder.

4.3 Where files land (data directory)

Step Default output path (under data dir)

```

-----
Gaia DR3 `GAIA_DR3/vyvar_gaia_dr3.db`
Blind indexes `GAIA_DR3/gaia_triangles_fine.pkl`, `gaia_triangles_wide.pkl`
VSX `VSX/vyvar_vsx_local.db`
Exoplanets `exoplanets/vyvar_exoplanet_local.db`

```

Override with script flags (``--out``, ``--db``, etc.) or set paths in Settings / ``config.json`` after building.

4.4 Time and disk (typical)

Step Network source Download Output size (typical)

Gaia G<=16.5 full sky esa.gaia.eu TAP hours to days **~9-10 GB** SQLite

Blind indexes (local Gaia DB) none ~100-500 MB PKL

VSX Vizier B/vsx/vsx minutes ~10-50 MB

Exoplanets exoplanetarchive.ipac.caltech.edu minutes ~1-5 MB

Gaia build is **resumable** (`strip_progress`` in the DB). Safe to restart after interruption. Narrow `--dec-min`/`--dec-max`` for testing before a full-sky run.

4.5 Verification

```

`./vyvar.sh --selftest`

`sqlite3 ~/.local/share/vyvar/GAIA_DR3/vyvar_gaia_dr3.db "SELECT COUNT(*) FROM gaia_dr3;"`

```

Row counts depend on mag limit and sky coverage. After building, confirm paths in Settings (Paths section) or `config.json`` point at the files above.

4.6 When to rebuild

Event Action

First install Full sequence (4.1)

Gaia mag limit raised Re-run Gaia + blind indexes

VSX mag limit raised Re-run VSX only (incremental)

Exoplanet archive update Re-run exoplanet builder (incremental)

New VYVAR install Keep data directory; rebuild only if you want fresh catalogs

5. Equipment setup (DB Explorer)

Open **Database Explorer** in the UI. Add:

1. **Location** (lat/lon/alt)
2. **Telescope**
3. **Equipment** (camera, filters, gain)

For **OSC cameras**, set **BAYERMASK** on the equipment record (required for correct debayer/channel handling).

6. Upgrade

1. Close VYVAR.
2. Replace the **install directory** with the new bundle (or unpack over it).
3. Do **not** delete the data directory - config, DB, and catalogs are preserved.

7. Uninstall

1. Delete the install directory.
2. Optionally delete the data directory if you no longer need observations/catalogs.

8. Troubleshooting

Issue Action

App will not start Run `--selftest`; check antivirus / Windows SmartScreen (unsigned binaries)

No stars / no comps Catalog paths empty or wrong - build catalogs (section 4)

Linux import errors Requires ****glibc >= 2.39**** (Ubuntu 24.04 baseline)

Wrong data location Set `VYVAR\_DATA\_DIR` before launching

License

See `LICENSE` in the install directory (proprietary; same terms as the dev repository).